

DSA Daily Quiz 3

Total points 9/10 ?

The respondent's email (**omjbhaleraowork@gmail.com**) was recorded on submission of this form.

Name *

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Last TWO Digits of PRN *

48



*1/1

What is the optimal time complexity to count the number of nodes in a linked list?

- ☐ $O(1)$
- ☒ $O(n)$
- ☐ $O(\log n)$
- ☐ none



✗ Which of these data types is used by operating system to manage the Recursion in Java? *0/1

- ☐ Array
- ☐ Stack
- ☐ Queue
- ☒ Tree

✗

Correct answer

- ☒ Stack



✓ What will be the output of the following program? *

1/1

```
class RecursiveFor
{
    public static void main(String[] args)
    {
        int out = 15;
        for (int i = 4; i < 6; i++)
        {
            for (int j = 7; j >= 5; j--)
            {
                if (i == j) continue;
                if (i > j)
                {
                    out += main(i, j);
                }
                else
                {
                    out += main(j, i);
                }
            }
        }
        System.out.println("out = " + out);
    }

    private static int main(int a, int b)
    {
        if (a + b == 0)
        {
            return 2;
        }
        return a + main(a - 1, b);
    }
}
```

- ☐ out = 78
- ☒ out = 100
- ☐ out = 80
- ☐ Compilation Error
- ☐ Goes into infinite loop



✓ What will be the output of the following program? *

1/1

```
class Recursion
{
    public static void main(String[] args)
    {
        int ma = 35;
        int in = 67;
        int sta = recurse(ma, in);
        System.out.println("static = " + sta + "tic");
    }

    public static int recurse(int pub, int lic)
    {
        if (pub % 13 == 0)
        {
            return pub + lic;
        }
        return recurse(++pub, lic / 2);
    }
}
```

- ☐ static = 41tic
- ☐ static = 47tic
- ☒ static = 43tic
- ☐ Compilation Error
- ☐ Throws StackOverflowError



✓ What's happen if base condition is not defined in recursion ? *

1/1

- ☐ Stack underflow
- ☒ Stack Overflow
- ☐ None of these
- ☐ Both a and b



✓ Select the correct output.

*

1/1

```
int rec(int num){  
    return (num) ? num%10 + rec(num/10):0;  
}
```

```
main(){
```

```
    System.out.println(rec(4567));  
}
```

- ☐ 4
- ☐ 12
- ☒ 22
- ☐ 21



✓ what will be the output of **func(3,8)** . *

1/1

```
int func(int a, int b){  
  
    if(b==0)  
        return 0;  
    if(b==1)  
        return a;  
  
    return a + func(a,b-1);  
  
}
```

- ☐ 11
- ☒ 24
- ☐ 22
- ☐ 21



✓ Choose the correct answer. *

1/1

- ☐ Recursion is always better than iteration.
- ☒ Recursion uses more memory compared to iteration. ✓
- ☐ Recursion uses less memory compared to iteration.
- ☐ Iterative function is always better and simpler to write than recursion.

✓ Which of the following problem cannot be solved using recursion? *

1/1

- ☐ Tower of Hanoi
- ☐ Fibonacci series
- ☐ Tree Traversal
- ☒ Problems without base case ✓

✓ To end a recursive method a RETURN statement is usually kept inside

*1/1

- ☐ IF block
- ☐ ELSE block
- ☒ IF or ELSE block ✓
- ☐ None

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